

Linear Regression

Linear Models, Least Squares, Inference, and
Diagnostics

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Session Roadmap

Recap & Objectives

Mathematical Prerequisites

Simple Linear Regression

Accuracy & Inference

Multiple Linear Regression

Diagnostics & Extensions

Applied View: Parametric vs Non-parametric

Summary



SECTION 1

Recap & Objectives

From the previous session

We framed supervised learning as estimating an unknown function $Y = f(\mathbf{x}) + \varepsilon$, and split the expected test error into (reducible) $\text{Bias}^2 + \text{Var} +$ (irreducible) σ^2 .

The gap this session fills

We now choose the simplest useful form for f : a linear one. It makes bias explicit, variance analysable in closed form, and every later method (ridge, lasso, GAMs, neural nets) a departure from this baseline.

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By the end of this session you will be able to:

1. Derive the least-squares estimator from first principles, in both scalar and matrix form.
2. Interpret coefficients, standard errors, confidence intervals and hypothesis tests.
3. Quantify model fit with **RSE**, R^2 and the **F-statistic**.
4. Diagnose the six classic failure modes and know their remedies.
5. Contrast linear regression with a non-parametric baseline (KNN).

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Why Linear Regression Is the Right First Model

Main idea

Linear regression models a quantitative response as a linear function of the input vector $\mathbf{x} = (x_1, \dots, x_p)^T$: it estimates the conditional mean $\mathbb{E}[Y \mid \mathbf{x}]$ by a linear approximation.

Why it matters

- ▶ Interpretable and cheap to fit.
- ▶ Gives coefficients, standard errors, intervals and tests.
- ▶ Foundation for GLMs, shrinkage, additive models, and beyond.

Model viewpoint

- ▶ Training set $\mathcal{T} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$.
- ▶ Response Y quantitative.
- ▶ Fitting criterion: minimise squared error.

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1. Vectors, inner products, norms

For $a, b \in \mathbb{R}^N$: $\langle a, b \rangle = a^\top b = \sum_i a_i b_i$, and the squared length $\|a\|^2 = a^\top a$.

2. Projection onto a subspace

The point in a subspace closest (in $\|\cdot\|$) to y is its orthogonal projection; the residual $y - \hat{y}$ is orthogonal to the subspace. This single fact *is* least squares.

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The Mathematics We Need First

3. Vector calculus: minimising a quadratic

For $g(\beta) = (y - X\beta)^\top (y - X\beta)$,

$$\nabla_{\beta} g = -2 X^\top (y - X\beta),$$

and setting $\nabla_{\beta} g = 0$ gives the minimiser (the function is convex).

4. Expectation and variance of estimators

$\mathbb{E}[\cdot]$ is linear; $\text{Var}(Az) = A \text{Var}(z) A^\top$. These turn the estimator into a random variable we can put error bars on.

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Setup and Population Model

Definition

Simple linear regression uses one input X to model the conditional mean of a quantitative response Y .

Population model

$$Y = \beta_0 + \beta_1 X + \varepsilon, \quad \mathbb{E}(\varepsilon) = 0, \quad \text{Var}(\varepsilon) = \sigma^2. \quad (3.1)$$

Symbol	Meaning
β_0	intercept: mean of Y at $X = 0$
β_1	slope: change in mean Y per unit X
ε	random error, mean 0, variance σ^2

From Prediction to a Loss Function

Prediction & residual

Estimating β by $\hat{\beta}$, the fitted value and residual are

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i, \quad e_i = y_i - \hat{y}_i.$$

Residual sum of squares — the loss

$$\text{RSS} = \sum_{i=1}^N e_i^2 = \sum_{i=1}^N (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2. \quad (3.3)$$

The idea in one line

“Training” means choosing $\hat{\beta}$ to minimise this loss — equivalently, making the residual vector as short as possible.

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Derivation: The Normal Equations (scalar)

Set the partial derivatives of $\text{RSS}(\beta_0, \beta_1)$ to zero:

$$\frac{\partial \text{RSS}}{\partial \beta_0} = -2 \sum_i (y_i - \beta_0 - \beta_1 x_i) = 0, \quad \frac{\partial \text{RSS}}{\partial \beta_1} = -2 \sum_i x_i (y_i - \beta_0 - \beta_1 x_i) = 0.$$

At the solution, $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$: the residual is orthogonal to both the intercept and predictor columns — exactly the projection picture from the prerequisites.

Closed-form estimates

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \quad (3.4)$$

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Least Squares in Matrix Form

Design-matrix formulation

Stack the data: $y = X\beta + \varepsilon$, with X the $N \times (p + 1)$ design matrix (first column all ones for the intercept).

Minimise the quadratic (prerequisite 3)

$$\text{RSS}(\beta) = (y - X\beta)^\top (y - X\beta) \implies X^\top X \hat{\beta} = X^\top y \quad (\text{normal equations}).$$

Closed form and the hat matrix

$$\hat{\beta} = (X^\top X)^{-1} X^\top y \quad \hat{y} = X\hat{\beta} = \underbrace{X(X^\top X)^{-1} X^\top}_H y.$$

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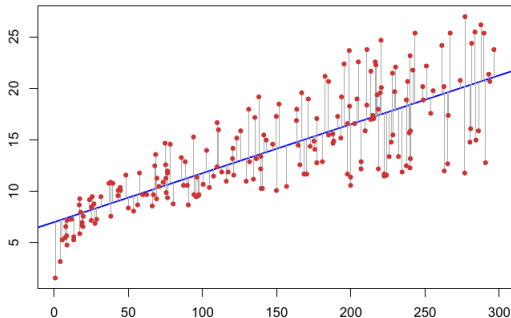
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Figure: The Least-Squares Fit



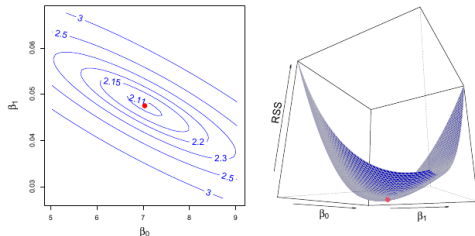
What you see

- ▶ Points: observed markets.
- ▶ Line: least-squares fit.
- ▶ Segments: residuals e_i .

Advertising fit

$\widehat{\text{sales}} = 7.03 + 0.0475 \times \text{TV}$: \$1,000 more TV spend $\approx$ 47.5 extra units.

Figure: RSS Is a Convex Bowl



Contours & surface

Each curve = constant RSS; the surface is bowl-shaped. Least squares sits at the unique bottom.

ML reading

RSS is the loss surface. Convexity is why the normal equations give a global optimum — no local minima to worry about.

How Uncertain Are the Coefficients?

Sampling distribution of least squares

Under $\text{Var}(\varepsilon) = \sigma^2 I$,

$$\mathbb{E}(\hat{\beta}) = \beta, \quad \text{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}.$$

Standard errors (simple regression)

$$\text{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}, \quad \text{SE}(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{N} + \frac{\bar{x}^2}{\sum_i (x_i - \bar{x})^2} \right].$$

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Confidence Intervals and the t -test

95% confidence interval

$$\hat{\beta}_1 \pm 2 \text{SE}(\hat{\beta}_1).$$

Advertising: $\beta_1 \in [0.042, 0.053]$.

Hypothesis test

$$H_0 : \beta_1 = 0 \text{ vs } H_a : \beta_1 \neq 0,$$

$$t = \frac{\hat{\beta}_1 - 0}{\text{SE}(\hat{\beta}_1)}. \quad (3.14)$$

Term	Coef.	Std. Err.	t	p -value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

Read it carefully

A tiny p -value is evidence of association *under the model* — not proof of causation.

Assessing Model Fit: RSE and R^2

Residual standard error

$$\text{RSE} = \sqrt{\frac{1}{N-2} \text{RSS}}. \quad (3.15)$$

Same units as Y ; “typical” error size.

Coefficient of determination

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}, \quad \text{TSS} = \sum_i (y_i - \bar{y})^2. \quad (3.17)$$

Fraction of variance explained.

TV-only Advertising model

$\text{RSE} = 3.26$, $R^2 = 0.612$: TV alone explains $\sim 61\%$ of the variation in sales. In simple regression, $R^2 = \text{Cor}(X, Y)^2$.

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From One Predictor to Many

General model

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon. \quad (3.19)$$

Same matrix solution as before: $\hat{\beta} = (X^T X)^{-1} X^T y$.

The key interpretive shift

β_j is the average effect of a one-unit increase in X_j , holding all other predictors fixed.
Equivalently (residualisation): the effect of the part of X_j not explained by the others.

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Why “Holding Others Fixed” Matters

Predictor	Coef.	Std. Err.	<i>t</i>	<i>p</i> -value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
Radio	0.189	0.0086	21.89	< 0.0001
Newspaper	−0.001	0.0059	−0.18	0.8599

The newspaper puzzle

Newspaper is significant alone but not after adjusting for TV and radio — because newspaper spend correlates with radio spend ($\text{Cor} = 0.35$). Simple regression let it take credit for radio's effect.

Is Any Predictor Useful? The F -test

Overall test

$H_0 : \beta_1 = \dots = \beta_p = 0$ versus “at least one $\neq 0$ ”:

$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(N - p - 1)}. \quad (3.23)$$

- ▶ If H_0 holds, $F \approx 1$; if some predictor helps, $F > 1$.
- ▶ Advertising (TV+radio+newspaper): $F = 570$ — overwhelming evidence.

Why not just scan individual p -values?

With many predictors some p -values are small by chance. The F -test accounts for p ; the partial F -test (with $F = t^2$ when $q = 1$) tests a subset.

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Fit Improves, Then Plateaus

Model	R^2	RSE
TV only	0.612	3.26
TV + radio + newspaper	0.897	1.69

Lesson

Adding radio to TV helps a lot; adding newspaper on top adds almost nothing. More predictors is not automatically better — a theme we return to in model selection.

Relaxing Additivity: Interactions

Add a product term

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon = \beta_0 + (\beta_1 + \beta_3 X_2) X_1 + \beta_2 X_2 + \varepsilon. \quad (3.31)$$

The effect of X_1 now depends on X_2 .

Advertising synergy

Adding TV $\times$ radio raises R^2 from 89.7% to 96.8%. Hierarchical principle: keep main effects whenever their interaction is in the model.

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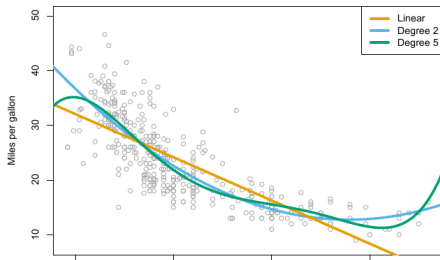
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Relaxing Linearity: Polynomial Terms



Auto data

$$\text{mpg} = \beta_0 + \beta_1 \text{hp} + \beta_2 \text{hp}^2 + \varepsilon.$$

Quadratic beats the straight line; degree-5 is needlessly wiggly.

Still “linear”

The model is linear *in the coefficients*, even though it curves in X . This trick underlies basis expansions later in the course.

The Six Things That Can Go Wrong

1. Non-linearity
2. Correlated errors
3. Non-constant variance
4. Outliers
5. High-leverage points
6. Collinearity

The discipline

A coefficient table is never enough. Read residual plots, leverage $h_i = H_{ii}$ (average $(p+1)/N$), and check collinearity before trusting a model.

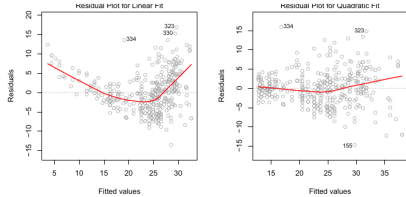
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Diagnosing Non-linearity and Heteroscedasticity



Residuals vs fitted: a U-shape signals missing curvature.

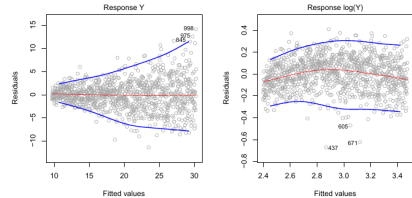


FIGURE 3.11. Residual plots. In each plot, the red line is a smooth fit to the residuals, intended to make it easier to identify a trend. The blue lines track the outer quantiles of the residuals, and emphasize patterns. Left: The funnel shape

A funnel signals non-constant variance; try $\log Y$ or WLS.

VIF

$$\text{VIF}(\hat{\beta}_j) = \frac{1}{1 - R_{X_j|X_{-j}}^2}.$$

Collinearity stretches the RSS valley: many coefficient pairs give near-equal RSS, so SEs balloon. Rule of thumb: $\text{VIF} > 5\text{--}10$ is trouble. Remedy: drop or combine predictors.

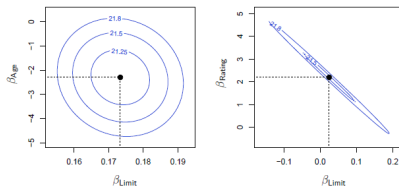


FIGURE 3.15. Contour plots for the RSS values as a function of the parameters β for various regressions involving the **Credit** data set. In each plot, the black dots represent the coefficient values corresponding to the minimum RSS. Left: A contour plot of RSS for the regression of **balance** onto **age** and **limit**. The minimum value is well defined. Right: A contour plot of RSS for the regression of **balance** onto **rating** and **limit**. Because of the collinearity, there are many pairs $(\beta_{\text{Limit}}, \beta_{\text{Rating}})$ with a similar value for RSS.

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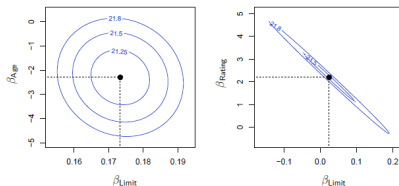


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SECTION 7

Applied View: Parametric vs Non-parametric

Linear Regression vs KNN Regression

Linear regression (parametric)

- ▶ Assumes a global linear form.
- ▶ Interpretable, testable, low variance.
- ▶ Biased if truth is very non-linear.

KNN regression (non-parametric)

$$\hat{f}(x_0) = \frac{1}{k} \sum_{x_i \in N_k(x_0)} y_i.$$

- ▶ Flexible, no global form.
- ▶ Low bias, higher variance.
- ▶ Suffers in high dimensions.

Curse of dimensionality

As p grows, “nearest” neighbours are far away, so local averaging fails. When data per predictor is scarce, the linear model often wins.

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- ▶ Biased if truth is very non-linear.

KNN regression (non-parametric)

$$\hat{f}(x_0) = \frac{1}{k} \sum_{x_i \in N_k(x_0)} y_i.$$

- ▶ Flexible, no global form.
- ▶ Low bias, higher variance.
- ▶ Suffers in high dimensions.

Curse of dimensionality

As p grows, “nearest” neighbours are far away, so local averaging fails. When data per predictor is scarce, the linear model often wins.

In Practice: Fitting in a Few Lines

```
import numpy as np
from sklearn.linear_model import (
    LinearRegression)

# X: (N,p) design, y: (N,)
model = LinearRegression().fit(X, y)
print(model.intercept_, model.coef_)

# closed form, for intuition:
Xb = np.c_[np.ones(len(X)), X]
beta = np.linalg.solve(Xb.T @ Xb, Xb.T @ y)
```

Same math, two doors

sklearn and the explicit normal-equation solve give identical $\hat{\beta}$. The library adds numerical stability (it uses a QR/SVD solve, not a raw inverse).

Habit

Always look at residual plots and R^2 on held-out data, not just the training fit.

Summary & What Comes Next

Key takeaways

- ▶ Least squares = orthogonal projection: $\hat{\beta} = (X^T X)^{-1} X^T y$.
- ▶ Inference (SE, CI, t , F) comes from $\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$.
- ▶ Fit with RSE / R^2 ; always run diagnostics before trusting it.

Next session: Classification

What if Y is a category, not a number? Squared error and a straight line no longer fit. We keep the linear predictor $\mathbf{x}^T \beta$ but pass it through a link function — leading to logistic regression.

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Linear Regression

From least squares to inference, diagnostics, and KNN

Questions & discussion welcome

